AZIMUTH BUSINESS ON WHEELS

DIGITAL TRANSFORMATION PROPOSAL

Azimuth BuildTrack

A single mobile platform to plan, track, and deliver every food truck & kiosk build — on time, with full traceability.

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Contents

01 Executive Summary

Azimuth Business on Wheels builds premium food trucks, carts, and kiosks. Each unit is a project that moves through roughly 17 stages — from the first sales call to final delivery — and at any given time the company runs **35+ projects in parallel**.

Today this is coordinated largely through people's memory, phone calls, and scattered records. That works until it doesn't. Two failure patterns repeatedly cost the company time, money, and client trust:

1. **Timeline derailment from missed lead times.** If equipment must be installed on the 17th, it has to be ordered by the 13th. Across 35 projects, these order-by dates get missed — and one miss pushes the whole project past its promised delivery date.
2. **No traceability of installed components.** When a part fails after delivery (recently a 42-inch TV), the team cannot quickly find that specific unit's bill and warranty — and cannot see which other trucks carry the same part.

This document proposes **Azimuth BuildTrack** — a single mobile application, used by every role in the company and by clients, that turns the entire build process into a structured, proactive, fully-recorded system.

> **THE CORE IDEA**
>
> Move from **reactive** (find out something was missed after it hurts) to **proactive** (the system warns the right person before it becomes a problem) — and make every truck a **digital twin** where every component, bill, and warranty is one tap away.

17	35+	1	2–3
build stages tracked	parallel projects managed	app for all roles	months to first working version

02 The Problem — Why This Matters

The build process itself is well understood. The problem is not *what* to do, but *keeping 35 of them coordinated at once* without anything slipping through the cracks. Two real, recent examples make it concrete.

PROBLEM 1 — A MISSED ORDER-BY DATE DERAILS A DELIVERY

Equipment was scheduled to be installed on the 17th. To make that date it needed to be ordered by the 13th. With dozens of projects running together, that order-by date was missed — so the equipment was not available on the 17th, and the entire project slipped past its promised delivery date.

PROBLEM 2 — A FAILED PART WITH NO TRACEABILITY

A delivered truck had a 42-inch TV that developed a fault. The same TV model had been installed in 25 other trucks. The team then had to manually hunt for the bill and warranty tied to that specific truck — a slow, error-prone effort — with no easy way to check which other trucks were exposed to the same risk.

The common root cause

Symptom	Root cause	What is missing
Orders placed too late deliveries slip	Lead times live in people's heads, not in a system	Proactive, automated scheduling & alerts
Cannot find a part's bill / warranty	No single record of what went into each truck	Per-unit component traceability
Clients call repeatedly for status	No transparent view of progress	A client-facing progress view
Delays happen but no one knows why	Reasons are verbal, never recorded	Structured delay reasons & data

03 Business Impact — The Cost of Inaction

Each of these issues carries a direct cost. Left unmanaged across 35+ parallel projects, they compound.

Area	What it costs today	What BuildTrack changes
Delivery delays	Penalties, idle workshop bays, delayed payments, damaged reputation	Early warnings prevent the miss before it happens
Warranty / service	Hours lost hunting bills; company may absorb costs that a vendor warranty would have covered	Bill & warranty for any part in seconds; claim on time
Recalls / defects	A bad batch stays hidden until units fail one by one	Instantly list every truck with the affected part
Client trust	Constant status calls; surprises about delays	Transparent progress builds confidence & referrals
Management time	Owners chase status manually across projects	One dashboard shows the health of all projects

> Even preventing **one** delayed delivery or recovering **one** warranty claim per month typically justifies the investment in the platform.

04 The Proposed Solution

Azimuth BuildTrack is one mobile app used by everyone involved in a build — internal teams and clients — each seeing only what their role needs. It digitises the full journey of every truck and actively manages it.

What the platform does, in one view

Capability	What it means for Azimuth
Structured project tracking	Every truck follows the 17-stage workflow with clear ownership and status
Proactive scheduling & alerts Hero 1	The system calculates order-by dates and warns before anything is missed
Component traceability Hero 2	Every part in every truck, with its bill and warranty, on record
Procurement management	Purchase orders, vendor lead times, and delivery tracking
Client progress view	Clients track their build without phone calls
Delay reasons & insights	Every slip is tagged with a reason, building data to fix root causes
Support & ticketing	Post-delivery issues raised and resolved in-app

DESIGN PRINCIPLE

The two "hero" capabilities directly solve the two real problems above. Everything else supports and extends them. We build the heroes first.

05 Hero Capability 1 — Scheduling & Alert Engine

Solves Problem 1: missed order-by dates that derail deliveries.

HOW IT WORKS

1. **Standard template.** Each of the 17 stages has a typical duration and its dependencies defined once (e.g. structure cladding electrical equipment install).
2. **Lead times per material.** Every item carries a known vendor lead time (e.g. chassis 20 days, kitchen equipment 6 days, TV 4 days).
3. **Backward scheduling (the key).** The system works backwards from each install date:

   Order-by date = Install date - Lead time - Safety buffer
   Example: install 17th - 3 days lead - 1 day buffer = **order by 13th**

4. **Proactive alerts.** Two days before the order-by date, the procurement owner is notified. If it is still not ordered, it escalates to the project manager and owner.
5. **Automatic cascade.** If any stage slips, all downstream dates and order-by dates recalculate automatically and everyone is re-alerted — no manual re-planning.
6. **Fleet health view.** One screen shows all 35 projects colour-coded: on-track, at-risk, delayed — with the most urgent actions on top.

Outcome: The "we forgot to order it in time" failure becomes structurally impossible — the system remembers, calculates, and warns for every project, every part, every day.

06 Hero Capability 2 — Component Traceability

Solves Problem 2: no record of what went into each truck.

Every physical truck becomes a **digital twin** — a complete record of every component installed in it. We track two distinct things:

Item Catalog (the type)

e.g. "Samsung 42-inch TV — Model UA42XYZ". A category of part.

Component Instance (the actual piece)

A specific physical unit with its serial number, purchase bill, vendor, warranty expiry, and which truck it went into, on what date, installed by whom.

The two questions it answers instantly

Question	Answer in the app
"What is installed in Truck #A12, and where is each bill/warranty?"	Full component list; tap any part for its bill, warranty, and vendor.
"This TV model is faulty — which other trucks have it?"	Every affected truck listed instantly proactive replacement / recall.

> **Bonus:** if a part is swapped during the build, the history is preserved too — you always know the final state *and* the full trail of what changed.

07 Delay Root-Cause Analysis & Solutions

Delays rarely have a single cause. Below is a structured map of why builds slip and exactly how the platform prevents or manages each. A foundational feature underpins all of them:

> **DELAY REASON CODES**
>
> Whenever a stage slips, the responsible person must tag *why* from a standard list (not free text). This makes the reason visible to the client instantly, and — over months — builds data that reveals the top systemic causes to fix.

1. Procurement / material delays

Reason	Solution in BuildTrack
Order forgotten (human error)	Backward-scheduled order-by alert + escalation
Vendor delivered late	Vendor reliability score auto-extends that vendor's lead time
Wrong / defective material received	Goods-receipt check with photo; mismatch flagged instantly
Item out of stock at vendor	"Confirm availability" step before order-by; alternate vendor
Partial delivery	PO marked partially received; shortfall auto-alerted
Imported item (customs)	Larger built-in lead-time buffer for such items

2. Dependency / sequencing

A stage cannot start until its predecessor finishes; one slip stalls the chain. **Solution:** dependency engine recalculates all downstream dates and highlights the critical path (the slip that actually moves delivery).

3. Design / approval delays

Clients take time to approve layouts or request revisions. **Solution:** in-app approval gate with reminders; change-orders show cost & timeline impact; time spent waiting on the client is logged to the client — protecting Azimuth from unfair blame.

4. Client-side delays

Reason	Solution
	Payment gate holds project start; automatic reminders

Advance payment not received work cannot start	
Milestone payment pending	Milestones linked to stages; optional work-hold flag
Indecision on branding / colour	In-app approval with deadline; delay logged to client

5. Workshop / resource delays

Limited bays and skilled staff (welders, electricians) create contention across 35 projects. **Solution:** capacity view (which truck in which bay), resource allocation (who is on what), and conflict flags for the PM to prioritise.

6. Quality / rework

Failed testing or a defective part on arrival. **Solution:** failed testing sends the stage into a rework loop with visible timeline impact; a defective part triggers an immediate warranty claim and replacement order via traceability.

7. Communication / handoff

The next person doesn't know it's their turn. **Solution:** stage completion auto-notifies the next owner; everything is logged — a single source of truth.

8. External factors

Reason	Solution
Weather (paint needs dry conditions)	Weather-dependent flag + buffer; reason tagged "weather"
Festivals / vendor closed	Holiday calendar built into scheduling
Transport / RTO registration / permits	Tracked as explicit tasks & compliance checklist

Above all — a "Risk Radar"

The platform warns *before* a delay, not just after. Each project shows a live health signal:

- **On-track** — all dates safe
- **At-risk** — an order-by date is near, or a stage is about to slip
- **Delayed** — a slip has occurred and delivery is affected

08 Users & Roles

One app, role-based views. Everyone sees only what they need; sensitive data (costs, payments) stays restricted.

Role	What they do in the app
Owner / Admin	Fleet-wide health, analytics, escalations, full visibility
Project Manager	Owns timelines across multiple projects; resolves conflicts
Procurement	Order-by alerts, purchase orders, vendors, lead times
Workshop / Fabrication	Update stage status, upload photos (works offline)
Design	Manage layout & design stage, client approvals
Sales	Leads, negotiation, contracts (later phase)
Store / Inventory	Serial numbers, bills, warranties, stock
Service / Support	Handle post-delivery tickets
Client	Track progress, approve designs, raise tickets

09 Full Module Map

The complete platform is made of the following modules. Not all ship at once — see the roadmap (Section 13).

Module	Purpose
Project Management	17-stage tracking, timelines, milestones
Scheduling & Alert Engine **Hero 1**	Order-by dates, cascade, risk radar
Component Traceability **Hero 2**	Serialized parts, bills, warranties, digital twin
Procurement & POs	Requisitions, vendor lead times, delivery tracking
Client Portal	Progress, photos, documents, ETA, delay reasons
Ticketing / Support	Post-delivery issues, SLA, warranty-linked
Notifications	Push, WhatsApp, SMS, email
Payments / Finance	Advance & milestone payments, invoices
CRM / Sales	Leads negotiation contract project
Analytics & Dashboards	Health, bottlenecks, vendor & failure insights

10 Client Experience

Giving clients their own view in the app reduces status calls and builds trust — a genuine differentiator for a premium brand.

- Live progress percentage and current stage of their truck
- Photo / video timeline of the build as it happens
- Expected delivery date — and, if it moves, the **reason** shown transparently
- All documents in one place: contract, invoices, warranties
- Design approvals directly in the app
- After delivery: raise a support ticket with a photo; track it to resolution

> When a delay is due to the client (e.g. slow approval or pending payment), that is shown honestly too — so expectations stay fair on both sides.

11 Technology Approach & Architecture

The platform is **mobile-first** (iOS + Android) with a secure cloud backend. Recommendations below are proven, scalable choices; final selection can flex to the team's skills.

Layer	Recommended	Why
Mobile app	Flutter	One codebase for iOS + Android; smooth, premium UI
Backend / API	Node.js (NestJS)	Structured, scalable; clean home for scheduling logic
Database	PostgreSQL	Relational data ideal for traceability & dependencies
Cache & jobs	Redis + queue	Fast date recalculation, alerts, daily risk radar
File storage	Cloud object storage	Bills, warranties, build photos
Notifications	Push + WhatsApp + SMS/ email	Reach the right person on the right channel
Security	Role-based access control	Each role sees only what it should

Key engineering choices

- **Offline-first for the workshop.** Weak network on the floor is expected; the app caches locally and syncs automatically when back online. Critical for adoption.
- **Background automation.** Scheduled jobs recalculate dates, run the daily risk radar, and send reminders without anyone triggering them.
- **Built to extend.** The AI layer plugs in later without rebuilding the system.

12 Artificial Intelligence Strategy

AI is introduced **deliberately and in stages**. The core value comes from reliable rules and good data — AI adds leverage once enough history has accumulated. Starting with AI on day one would mean having no data to learn from.

Level	When	What it does
Statistics (not "AI", but feels like it)	Early	Auto-tune lead times & stage durations from actuals; vendor reliability scores
Predictive ML	After ~30–50 completed projects	Predict delay risk per project; suggest optimal ordering & resource allocation
Generative AI (LLM)	Experience polish	Client chatbot ("when will my truck be ready?"), automatic ticket triage, owner daily summaries

> **Honest position:** the MVP does not need AI. We architect for it now, switch it on when the data makes it genuinely valuable.

13 Phased Delivery Roadmap

Phase 1	Phase 2	Phase 3	Phase 4	Phase 5
Core	Client	Support & Ops	Sales & Money	AI

Phase 1 — Core Build Management MVP

Solves both real problems and gives a usable app.

- Role-based mobile app (workshop, PM, procurement, admin, basic client)
- Project + 17-stage tracking with timeline
- Scheduling & order-by alert engine (Hero 1)
- Procurement / PO tracking with vendor lead times
- Component traceability with bills & warranty (Hero 2)
- Push notifications, delay reason codes, offline sync
- Basic client progress view (progress + ETA)

Phase 2 — Client Experience & Communication

- Full client portal (photos/video, documents, detailed delay reasons)
- WhatsApp + SMS + email updates
- Design/layout approval flow; change-order management

Phase 3 — Support & Operations Depth

- Ticketing / support system (warranty-linked)
- Recall / batch management (built on traceability)
- Workshop capacity & resource planning
- Full vendor management

Phase 4 — Business & Money

- CRM / sales funnel (lead   contract   project)
- Payments / finance, payment gateway, accounting export

Phase 5 — Intelligence (AI)

- Predictive delay risk; smart scheduling suggestions
- Client chatbot; ticket auto-triage; owner daily summaries

> Phase order is not fixed. If post-delivery complaints are the bigger pain, ticketing can move earlier. We finalise the order together based on priorities.

14 Timeline & Team

An honest view, so expectations are set correctly:

> **The complete platform** (all modules + AI) is realistically a **6–9 month** effort. A focused **Phase 1 MVP** that solves both core problems is achievable in **2–3 months** with a small dedicated team.

Indicative Phase 1 plan (~10 weeks)

Weeks	Focus
1–2	Data model, authentication, roles, app skeleton
3–5	Project / stage tracking + scheduling engine
6–7	Procurement + component traceability
8–9	Alerts / notifications, delay reasons, basic client view
10	Testing, offline sync, polish, pilot on 2–3 live projects

Important: this timeline depends on team size. A single developer roughly doubles it; a small team of 3–4 makes the above realistic. Team size is being confirmed and the plan will be finalised accordingly.

15 Success Metrics (KPIs)

How we will measure whether the platform is working:

Metric	Target direction
On-time delivery rate	Up
Number of missed order-by dates	Down to near zero
Time to find any part's bill / warranty	From hours to seconds
Warranty claims recovered from vendors	Up
Client status phone calls	Down
Average delay per project	Down
Projects visible as "at-risk" before they slip	Up (earlier warning)

16 Decisions Needed & Next Steps

Inputs needed to finalise the plan

1. **Team size** — how many developers will build this (drives the timeline).
2. **Average build time** per truck (calibrates the scheduling templates).
3. **Vendors** — fixed set, or different per project (shapes procurement).
4. **Client visibility** — how much to show (full delay reasons vs high-level).
5. **MVP priority** — both hero features together, or one first.
6. **Existing tools** — anything in use today (Excel, WhatsApp, software) to migrate from.

Proposed next steps

1. Align on this approach and confirm the Phase 1 scope.
2. Confirm team size and lock the delivery timeline.
3. Design the detailed screens and data model for Phase 1.
4. Build the MVP and pilot it on 2–3 live projects.
5. Roll out to all projects, then proceed phase by phase.

> **In one line:** BuildTrack turns 35 parallel builds from something held together by memory and phone calls into a single, proactive system — so deliveries stay on time and every part is accounted for.

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